



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-2027
Course Handout (Part II)

Date: 31/07/2026

In addition to Part I (General Handout for all courses appended to the Timetable), this portion provide further details regarding the course.

Course No. : MATH F211

Course Title : **Differential Equations/ Mathematics-III**

Instructor In-charge : **KRISHNENDRA SHEKHAWAT**
(6067Q, Department of Mathematics)

Instructor(s) : Ankan Pal , Ashish Tiwari, Aritra Banerjee, Chandra Shekhar, Deepak Bhoirya, Pramod Eyyunni, Rakhee, Rijubrata Kundu, Shailesh Trivedi, Rishikesh Vaidya, Sudarshan Santra

1. Course Description

This course presents the study of differential equations with the objective of introducing the classical methods for solving boundary value problems. It also serves as a basis for the applications of differential equations, Fourier series and Laplace transform in the various branches of engineering and sciences. Furthermore, it emphasizes on the role of orthogonal polynomials in dealing with Sturm-Liouville problems.

2. Scope and Objectives

- To understand the theory of first and second order ordinary differential equations and to learn methods to solve them.
- To understand the series solution of second order ordinary differential equations and their interval of convergence.
- To understand the Laplace transform, its properties, and applications for solving IVPs and BVPs.
- To understand the Fourier series expansion of a function.
- To understand the concept of eigenvalues and eigenfunctions and to use them in solving heat and wave equations.

3. COURSE LEARNING OUTCOMES: After completing this course, the student will be able to

- 1) solve differential equations appearing in modeling of various processes in their respective engineering or science stream.
- 2) understand whether a given initial value problem is solvable or not.
- 3) understand the region in which the solution is valid (like in series solution approach).
- 4) solve the problems involving linear system of first order differential equations.
- 5) understand the eigenfunction expansion/Bessel series expansion of a function using respective orthogonality properties which can be further helpful in solving various problems in their engineering and science streams.





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4. Textbook

- i. Simmons G.F., Differential Equations with Applications and Historical Notes, Tata McGraw Hill, 2nd ed., 1991.

Reference Books:

- i. Shepley L. Ross: Differential Equations, John Willy & Sons, 3rd ed., 1984.
- ii. Edwards & Penney: Differential Equation and Boundary Value Problems, Pearson Education, 3rd ed., 2009.
- iii. W.E. Boyce & R. C. DiPrima: Elementary Differential Equations and Boundary Value Problems, John Wiley & Sons, Inc., 9th ed., 2009.





5. Course Plan

Week No.	Module Number	Lecture session	Section	Learning Outcome
1	1. To introduce classical methods for solving the first order ordinary differential Equations (L1 to L4)	Revision and self-study	1-6	Students will be able to solve the first order differential equations of various types.
		L1.1 First order equations	7	
		L1.2 Exact differential equations	8, 9	
		L1.3 Linear differential equation	10	
		L1.4 Reduction of order	11	
2-3	2. To introduce the classical methods to solve second order ordinary differential equations (L5 to L9)	L2.1-2.2 General solutions of second order results on linearly dependent (L.D.) and	14-15	Students will be able to the homogeneous order ordinary differential equations along with the knowledge of theoretical differential equations.
		L2.3 Use of a known solution to determine another L.I. solution of a differential	16	
		L2.4-2.5 Solving the second order homogeneous linear ordinary equations	17	
4-5	3. To obtain the particular solution of non-homogeneous second order ordinary differential Equations (L10 to L13)	L3.1-L3.2 Method of undetermined coefficients	18	Students will be able to obtain a particular solution of the non-homogeneous second order linear ordinary differential equations
		L3.3-L3.4 Method of variation of	19	
		Operator method (self study)	23	
6	4. To discuss the properties of solutions(L14to L16)	L4.1 Oscillations, Sturm Separation theorem and theorem on infinitely many positive zeros of solution	24	Students will be able to the behavior of solution second order differential without solving them.
		L4.2-L4.3 Sturm comparison theorem	25	
7-8	5. To introduce series solutions method to solve the second order linear differential equation	L5.1-5.2. Review of power series and series solution of first order ordinary differential equation, series solutions about	26, 27, 28	Students will be able to the second order differential equations with variable coefficients by using series solution approach.
		L5.3-5.4 Series solutions about regular	29, 30	



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	with variable coefficients (L17 to L22)	L5.5-5.6 Hypergeometric equations	31	
9-10	6. Special functions arising from series solutions of the second order linear ordinary differential equation (L23 to L28)	L6.1-6.6 Legendre polynomials, Bessel functions	44, 45, 46, 47	Students will be able to apply properties of Legendre polynomials and Bessel functions to solve various initial/boundary value problems in their respective engineering and science streams.
11-12	7. Introduction to Laplace Transform, its properties and applications (L29 to L34)	L7.1-7.2 Laplace transform and its existence, inverse Laplace transform	48,49	Students will be able to use the properties of Laplace transform in solving initial value problems and boundary value problems.
		L7.3 Applications to differential equations	50	
		L7.4 Derivatives and integrals of Laplace transforms	51	
		L7.5-7.6 Convolutions	52	
	8. To introduce system of first order ordinary differential equations. (self study)	Theory on system of equations and introduction to linear system	54, 55	Students will be able to solve the linear system of first order differential equations with the knowledge of theory of system of first order differential equations.
		Homogeneous linear systems with constant coefficients	56	
13	9. To introduce Fourier series (L35 to L37)	L 9.1 Fourier coefficients and problem of convergence	33-34	Students will be able to obtain Fourier series expansion of functions in a given interval, to know its convergence and apply it in their respective streams such as getting solution of a vibrating string etc.
		L9.2 Even and odd functions, cosine and sine series	35	
		L9.3 Extension to arbitrary intervals	36	
14	10. Boundary value problems. (L38)	L10.1 Eigenvalues and eigenfunctions	40	Students will be able to solve boundary value problems and to obtain eigenfunction expansion of a function using the orthogonality property of eigenfunctions of Sturm Liouville problems.





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14	11.To introduce separation of variables method to solve partial differential equations.(L39-L40)	L11.1. One dimensional wave equation	40	Students will be able to use separation of variable technique for solving some partial differential equations.
		L11.2 One dimensional heat equation	41	
		Laplace equation (self-study)	42	

***In common hours, practice problems will be done on topics covered in previous lectures.**

7. Evaluation Scheme:

	Evaluation Components	Weightage (Marks)
1	Mid Semester Exam	35% (CB)
2	Quizzes (Announced, best score of 2 quizzes will be considered out of 3 quizzes)**	20% (CB)
3	Comprehensive Exam	45% (CB/OB)

**** No make-up will be given under any circumstances**

8. Cut off for NC grade: Below 30 marks out of 300

9. Chamber consultation hour: To be announced in the class.

10. Notices: All notices regarding MATH F211 will be displayed on NALANDA.

Note: It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend lectures and common hours as per the schedule mentioned in time-table. Quizzes, tests and comprehensive examination are according to the evaluation scheme given in the respective course handout. If the student is unable to appear for the regular test/examination due to genuine exigencies, the student must refer to the procedure for applying for make-up test/examination.

KRISHNENDRA SHEKHAWAT(Instructor in charge)





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